HR ATTRITION ANALYTICS REPORT

Executive Summary

Employee attrition remains a critical challenge impacting organizational productivity, hiring costs, and workforce stability. This project leverages machine learning and business intelligence to identify the key drivers of employee resignation and proactively predict employees at risk of leaving.

Using historical HR data, a predictive model was developed and integrated with an interactive Power BI dashboard to support data-driven retention strategies. The analysis reveals that overtime workload, compensation levels, and job role pressures are the primary contributors to attrition risk.

The final solution enables HR teams to move from reactive hiring to proactive retention management.

Project Objectives

- Analyze historical HR data to understand attrition patterns
- Identify key factors influencing employee resignation
- Build a predictive model to forecast attrition risk
- Segment employees into actionable risk categories
- Develop an executive Power BI dashboard for monitoring
- Provide strategic recommendations to reduce attrition

Methodology

Data Preparation & EDA

The dataset was cleaned and explored using Python (Pandas, Seaborn). Key analyses included:

- Department-wise attrition distribution
- Salary band impact
- Overtime vs attrition relationship
- Promotion and tenure analysis
- Risk segmentation using predicted probabilities

Exploratory analysis indicated strong correlation between overtime workload and employee exits.

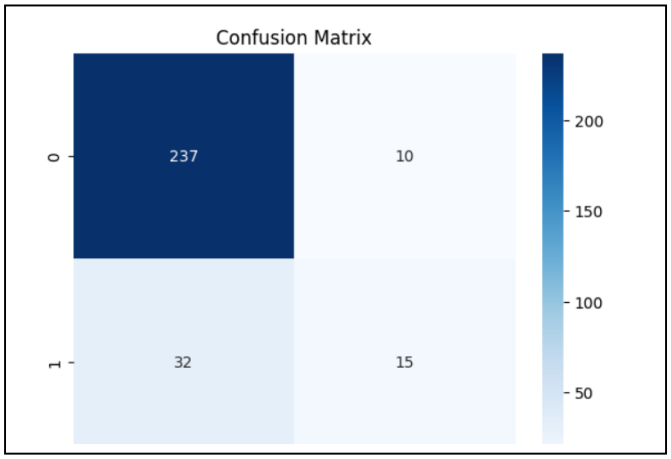
Predictive Modelling

Multiple classification models were evaluated:

- Logistic Regression (balanced)
- Logistic Regression with SMOTE
- Random Forest Classifier

The Random Forest model delivered the best performance and was selected as the final model.

Model Performance



Metric	Random Forest
Accuracy	0.857
Precision	0.83
Recall	0.11
F1-Score	0.19
ROC-AUC	0.812

Interpretation:

- The model effectively identifies employees likely to stay
- Recall for attrition cases is relatively low
- Model is suitable for **early risk flagging and prioritization**

Key Risk Drivers (SHAP Insights)

SHAP analysis identified the most influential factors driving employee attrition:

- Overtime workload
- Monthly income
- Years at company without promotion
- Job role pressure
- Work-life balance indicators

Employees working frequent overtime show significantly higher attrition probability.

Dashboard Highlights

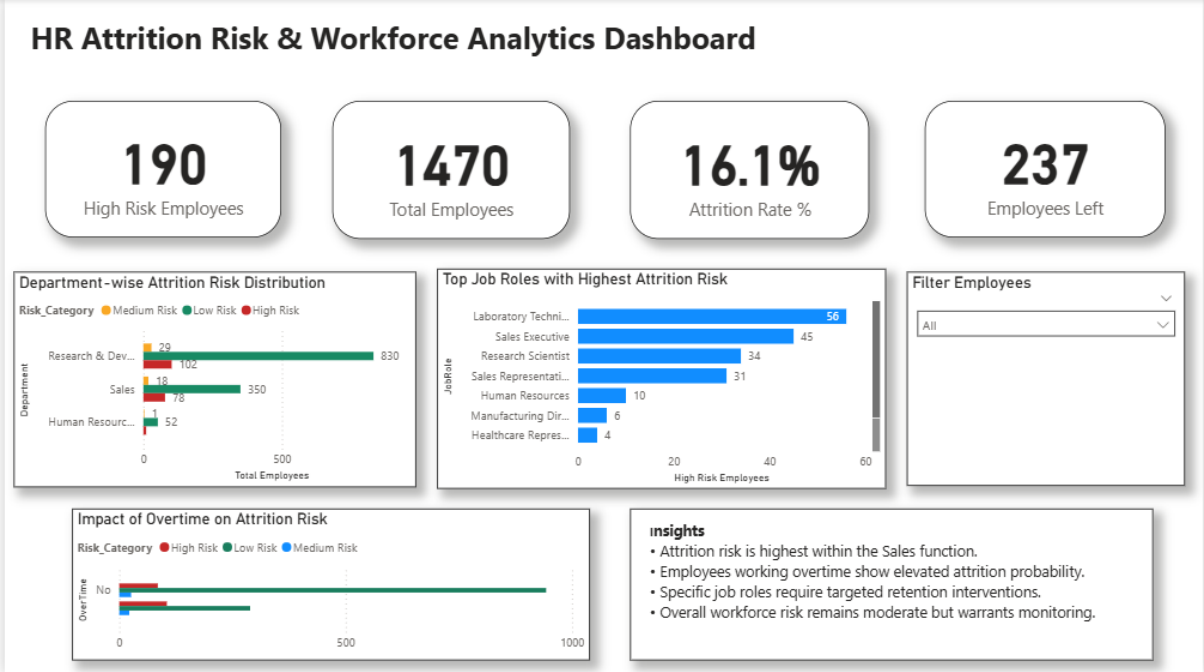
The Power BI dashboard provides real-time workforce intelligence:

Key metrics monitored:

- Total Employees: 1470
- High-Risk Employees: 190
- Attrition Rate: 16.1%
- Employees Left: 237

Major observations:

- Sales department shows highest attrition risk
- Overtime employees have elevated risk levels
- Certain job roles require immediate HR attention
- Overall workforce risk is moderate but rising



Recommended HR Actions

1. Compensation Optimization

- Review salary competitiveness for high-risk roles
- Introduce performance-linked incentives
- Benchmark compensation against industry standards

2. Workload Management

- Monitor and reduce excessive overtime
- Improve workforce planning
- Introduce flexible work arrangements

3. Career Growth Programs

- Establish transparent promotion pathways
- Conduct quarterly career discussions
- Implement internal mobility programs

4. Targeted Retention Strategy

For employees flagged as **High Risk (≥ 0.6 probability):**

- Conduct stay interviews
- Provide personalized development plans
- Review compensation positioning
- Manager intervention and engagement check

Monitoring & Future Enhancements

To sustain impact, the organization should:

- Review attrition dashboard monthly
- Retrain the model quarterly
- Implement early warning alerts
- Track department-wise risk trends
- Integrate employee engagement survey data

Expected Impact: Potential reduction of attrition by **15–25% annually** with proactive interventions.

Conclusion

The HR Attrition Analytics solution successfully demonstrates how predictive modeling and explainable AI can enhance workforce management. The Random Forest model provides reliable early warning signals, while SHAP analysis improves transparency of the key attrition drivers. Combined with the Power BI dashboard, the framework enables HR teams to proactively identify high-risk employees and implement targeted retention strategies. With continuous monitoring and model retraining, the organization can significantly reduce attrition and improve workforce stability.